

BOS Genesis MoP Creation Agent

Professional Release Notes and Package Intelligence Report

Version 0.1.0 | Public GitHub repository analysis | Generated 05-Jun-2026

13

PUBLIC TAGS

14

COMMITTS

0.1.0

PACKAGE VERSION

Py 3.11+

RUNTIME

Release positioning
This document packages the visible repository evolution into an executive-ready release note with change history from the first tags, package metadata, capability analysis, deployment posture, safety notes, validation checklist, risks, and next-release recommendations.

Audience	Purpose
Engineering Leaders	Understand release maturity, delivery velocity, and technical scope.
DevOps / Platform	Review packaging, Kubernetes/Helm posture, validation, and rollback readiness.
Citizen Development / Agent Platform Teams	Understand how this agent contributes to reproducible BOS Genesis installation documentation.

1. Executive Summary

Commercial summary of the v0.1.0 package, based on public repository evidence.

The BOS Genesis MoP Creation Agent is positioned as a spec-first, LLM-assisted service for reconstructing how a single BOS Genesis Kubernetes namespace was installed and for generating reproducible installation documentation. The README identifies two primary outputs: a human-readable Method of Procedure PDF and agent-readable Markdown installation notes.

The current public repository describes a foundation release with FastAPI service shell, REST endpoints, MCP-style tools, YAML/environment configuration loading, JSON structured logging, Dockerfile, Helm skeleton, deployment playbooks, optional ingress, pytest, and Ruff CI readiness. The README also states that Phase 1 generation returns IDs, trace placeholders, and artifact paths only, and does not yet call Kubernetes, Helm, Qdrant, or datastore integrations.

11 days

Release window

13

Visible tags

97.4% Py

Codebase mix

Read-only

Safety mode

Spec-first operating model	MCP-style agent contract	REST-triggered generation	Human MoP PDF target
Markdown installation notes target	Read-only safety posture	Helm/Kubernetes deployment path	LangGraph/LangMem-ready architecture

Release recommendation

This release can be presented commercially as an initial platform-capability release, not as a production-complete autonomous MoP generator. The strongest story is that the repo has moved beyond a blank scaffold into a tagged, incrementally delivered agent package with API/MCP contracts, document-rendering milestones, reasoning and memory milestones, and explicit safety constraints.

2. Release and Package Information

Canonical package, source, and deployment metadata for release documentation.

Field	Value
Product / Component	BOS Genesis MoP Creation Agent
Repository	github.com/aveeshek/bosgenesis-mop-creation-agent
Package name	bosgenesis-mop-creation-agent
Current package version	0.1.0
Release type	Foundation / initial platform capability release
Runtime requirement	Python >= 3.11
Primary interface	FastAPI REST service plus MCP-style HTTP contract
Primary target namespace	bosgenesis
Target deployment model	Docker image + Helm chart + Kubernetes namespace deployment
Security posture	Read-only discovery and document generation; no infrastructure mutation

Runtime dependencies from package metadata

fastapi	uvicorn	pydantic	PyYAML
mcp	langgraph	langchain-openai	langchain-ollama
redis	psycopg[binary]	clickhouse-connect	azure-identity

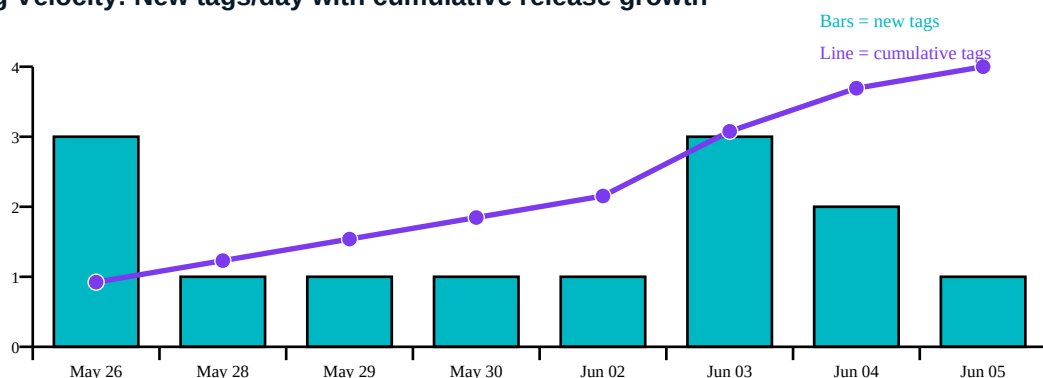
Dev and quality dependencies

pytest	httpx	ruff
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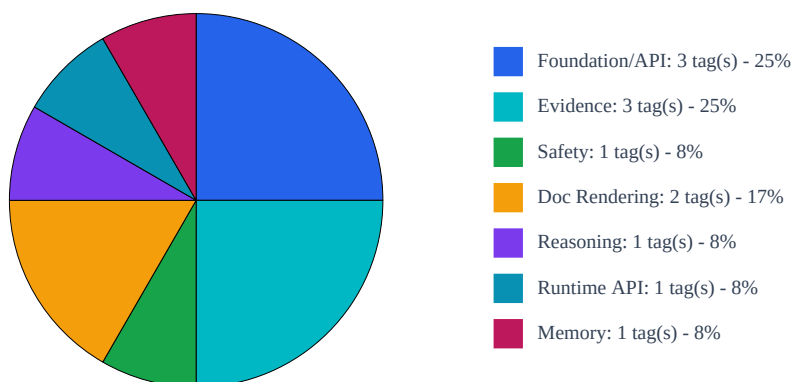
3. Analytical Release Dashboard

Delivery velocity, capability mix, and maturity signals inferred from public repo evidence.

Tag Velocity: New tags/day with cumulative release growth



Capability Distribution by Tag Theme



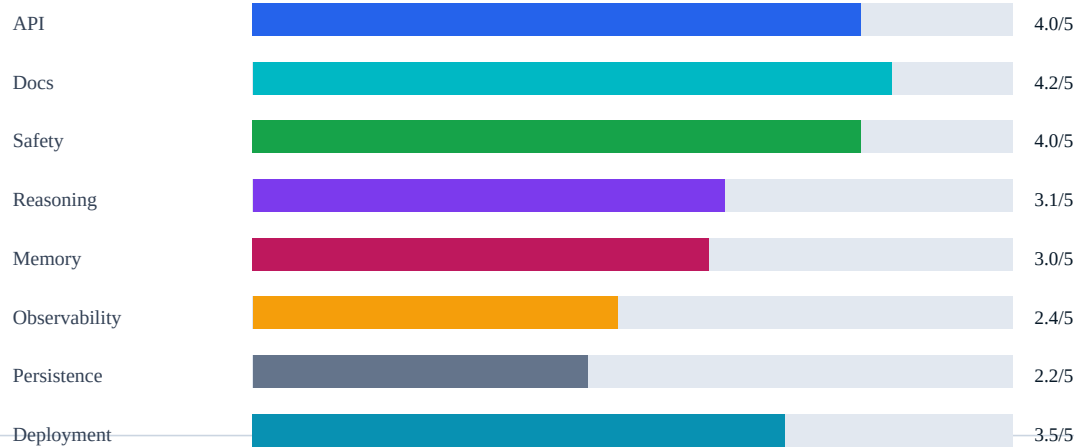
Note: classification inferred from public tag names and commit messages; not a formal GitHub release taxonomy.

4. Release Readiness Assessment

Analytical maturity scoring; intended for planning and review, not as a formal audit.

Release Readiness Assessment (Analytical Estimate)

Score based on public repo evidence: contracts, tags, dependencies, docs, and implementation statements.



Dimension	Score	Evidence-based interpretation	Recommended action
API contract	4.0/5	REST and MCP-style contracts are visible and testable.	Add formal OpenAPI examples and backward compatibility policy.
Document generation	4.2/5	PDF and Markdown renderer milestones are tagged.	Add golden sample artifact comparison tests.
Safety / governance	4.0/5	Explicit read-only and no-mutation safety posture.	Add policy-as-code tests for forbidden operations.
Reasoning workflow	3.1/5	Bounded LLM reasoning is tagged and dependencies are present.	Add deterministic evaluation suite for reasoning decisions.
Memory layer	3.0/5	Memory layer milestone exists with LangMem/Redis/Postgres direction.	Document memory retention, redaction, and reset semantics.
Observability	2.4/5	Docs mention Langfuse, SigNoz, OpenTelemetry, but current visibility is unclear.	Add trace IDs, spans, metrics, and dashboard screenshots.
Persistence	2.2/5	Package lists psycpg and storage roadmap, but durability status is not fully clear from public docs.	Add run metadata schema and migration scripts.
Deployment packaging	3.5/5	Dockerfile, Helm chart skeleton and scripts are visible in README.	Add production values, probes, resource limits, and release pipeline artifacts.

5. Change History from First Tags

Complete visible tag timeline converted into business-friendly release history.

The table below starts from the earliest visible tags and moves through the latest visible tag. Dates, commit hashes, and tag names are taken from the public GitHub tags page; release themes and value summaries are derived from the tag names and commit messages.

Tag	Date	Commit	Release theme	Commercial value / change summary
phase0-foundation	May 26, 2026	e74d61b	Foundation	Initial service foundation; packaging, config and repository skeleton for MoP Creation Agent.
phase1-mcp-api	May 26, 2026	2761063	MCP/API Contract	MCP-style contract and REST access path for external agent integration.
specs	May 26, 2026	571cc8d	Specifications	Spec-first design baseline and governance contract for future implementation.
phase03-snapshot-reader	May 28, 2026	9adeec3	Evidence Input	Snapshot reader capability for namespace evidence ingestion path.
phase04-mcp-enrichment	May 29, 2026	15765fa	MCP Enrichment	Enrichment route for evidence and tool-driven context expansion.
phase05-classification-and-safety	May 30, 2026	bd9510e	Safety/Classification	Safety and classification layer for bounded, read-only MoP generation behavior.
phase06-manifest-and-helm-reconstruction	Jun 02, 2026	6a2b4f1	Helm/Manifest Reconstruction	Reconstruct Kubernetes manifests and Helm-based install provenance.
phase08-markdown-installation-notes-renderer	Jun 03, 2026	2d69e25	Agent Notes Renderer	Render agent-readable Markdown installation notes for future execution.
phase09-qdrant-read-only-reference-lookup	Jun 03, 2026	9c9313e	Knowledge Lookup	Read-only Qdrant lookup path for reference/knowledge grounding.
phase10-llm-reasoning	Jun 03, 2026	e680d05	LLM Reasoning	Bounded LLM reasoning workflow for MoP creation decisions.
phase07-professional-pdf-renderer	Jun 04, 2026	f7c5f91	PDF Renderer	Professional PDF renderer for human-readable MoP output.
phase10.1-runtime-namespace-api	Jun 04, 2026	5fc14e1	Runtime Namespace API	Runtime APIs for target namespace discovery/generation interaction.
phase11-memory-layer	Jun 05, 2026	caf175b	Memory Layer	Memory layer for improved iterative MoP creation and context continuity.

6. Technical Capability Map

What the release enables today and what remains clearly marked as future or partial scope.

Capability area	Delivered / visible in repo evidence	Commercial interpretation
REST service shell	Health, config, generate, get-by-id, latest endpoints described in README.	Establishes service interface for demos, smoke tests, and integration discussion.
MCP-style integration	Health, generate, get, latest, and effective-config tools described.	Allows Codex or MCP-aware clients to call bounded MoP operations.
Document outputs	MoP PDF and Markdown installation notes are the stated primary outputs.	Creates dual-persona documentation: human operations and future agent execution.
Kubernetes/Helm reconstruction	Manifest and Helm reconstruction tag exists; SPEC scope is namespace-only.	Aligns directly with BOS Genesis reproducibility and environment reconstruction use case.
Qdrant lookup	Read-only reference lookup tag exists.	Supports grounded reference lookup without mutating knowledge stores.
LLM reasoning	Bounded LLM reasoning tag and LangGraph/LangChain dependencies exist.	Positions the component as an agentic assistant, not only a static renderer.
Memory layer	Memory tag plus LangMem/Redis/Postgres design direction.	Supports long-running MoP refinement and context continuity roadmap.
Safety posture	Read-only, no data export, no infrastructure mutation.	Strong fit for enterprise controlled automation and audit requirements.

Non-functional notes

Spec-driven implementation	Read-only discovery	No secret exposure	Traceability-oriented design
Kubernetes namespace boundary	Helm-centric reconstruction	Artifact-oriented output	Future observability hooks

7. Deployment, Validation, and Rollback Notes

Operator-ready release details for a professional deployment note.

The README describes local execution through editable Python install and an entrypoint command. It also describes Helm deployment using IMAGE_REPOSITORY and IMAGE_TAG, plus optional INGRESS_HOST for cluster exposure.

Purpose	Command
Local install	python -m pip install -e "[dev]"
Local run	bosgenesis-mop-creation-agent
Helm deploy	IMAGE_REPOSITORY=/bosgenesis-mop-creation-agent IMAGE_TAG= ./playbook/deploy.sh
Custom ingress	INGRESS_HOST=mop-creation-agent.bosgenesis.local IMAGE_REPOSITORY=/bosgenesis-mop-creation-agent IMAGE_TAG= ./playbook/deploy.sh
REST contract test	curl -X POST http://localhost:8080/mop-creation/generate -H "Content-Type: application/json" -d {target_namespace...}
MCP-style test	curl http://localhost:8080/mcp/tools; then POST tools/call for mop_creation_health

Recommended production-readiness checklist

Gate	Status to validate before promotion
Build integrity	Docker image built, signed/scanned, and mapped to exact Git tag/commit.
API health	/health returns healthy status after Helm deployment.
Config safety	/config/effective redacts all sensitive values.
MCP contract	Tool list and mop_creation_health tool call succeed.
Artifact output	PDF and Markdown output paths are created, validated, and retained.
Safety test	Agent cannot mutate Kubernetes, Helm, databases, streams, or application data.
Observability	Correlation IDs, logs, traces, and metrics visible in target tools.
Rollback	Uninstaller and Helm rollback path tested in non-production namespace.

8. Known Limitations, Risks, and Controls

Transparent release note section for commercial-grade governance.

Risk / limitation	Impact	Recommended control
Generation behavior is still partially contract/stub-oriented in README.	Users may expect full cluster/Helm inspection before it is complete.	Mark release as foundation capability; label production readiness clearly.
No full public release notes page appears on GitHub; tags are the primary change history.	Consumers may lack detailed feature-by-feature release explanation.	Publish formal GitHub releases for each major phase.
Observability integrations are described but current completeness is not fully visible from public docs.	Harder to prove runtime traceability in customer demo.	Include screenshots and sample trace IDs in next release.
Memory and LLM reasoning require strong guardrails.	Risk of nondeterministic or overreaching recommendations.	Add bounded prompts, policy checks, and regression evaluations.
Kubernetes/Helm reconstruction must avoid secrets and mutation.	Potential compliance risk if evidence export is too broad.	Namespace-only RBAC, secret redaction, deny-list and audit events.

Release classification

Classification	Recommended wording
External demo	Safe for controlled demo as an initial agent package and release-history example.
Internal platform pilot	Suitable for sandbox/pilot validation where read-only boundaries and artifact outputs are verified.
Production automation	Not yet recommended unless full evidence collection, artifact generation, security, and observability are validated.

9. Recommended Next Release Scope

Commercially useful roadmap derived from gaps and existing tag direction.

Priority	Theme	Recommended next capability	Business value
P1	Formal release management	Create GitHub Releases with release notes, binaries/artifact links, SBOM and checksums.	Improves commercial credibility and traceable delivery.
P1	Evidence engine	Complete namespace-scoped Kubernetes and Helm evidence collection.	Makes generated MoP grounded in runtime facts.
P1	Artifact generation	Generate real MoP PDF and Markdown notes with deterministic templates.	Turns the agent from contract demo into usable documentation factory.
P1	Security	Add redaction, secret deny-list, read-only RBAC and policy tests.	Makes enterprise adoption safer.
P2	Observability	Add OpenTelemetry spans, Langfuse traces, SigNoz dashboards and correlation IDs.	Improves supportability and executive demo evidence.
P2	Persistence	Store MoP runs, artifacts, evidence bundles, decisions and audit records in Postgres/MongoDB.	Supports history, audit, and repeatability.
P2	Evaluation	Add reproducibility, grounding, safety and PDF quality evaluation suites.	Improves confidence before promotion.
P3	Agentic memory	Add memory retention/reset semantics and context summarization.	Supports long-running refinement without stale context risk.

Suggested next release title

v0.2.0 - Evidence-Grounded MoP Artifact Generation: complete Kubernetes/Helm evidence extraction, deterministic PDF/Markdown generation, artifact retention, and smoke-test verification.

Appendix A - Evidence Sources and Assumptions

Source traceability for the sample release-note report.

Evidence area	Public repo evidence used
Repository status	Public GitHub repository; main branch; 14 commits; 13 tags; 0 stars/forks at retrieval time.
README intent	Spec-first, LLM-assisted agent to reconstruct one BOS Genesis namespace and generate reproducible installation documentation.
Primary outputs	Human-readable MoP PDF and LLM/agent-readable Markdown installation notes.
Runtime forms	On-demand MCP server integrated with Codex and standalone REST-triggered agent using external LLM and memory.
Safety posture	Read-only discovery/document generation; must not populate data, expose secrets, or mutate infrastructure.
Package metadata	Python >=3.11; project version 0.1.0; FastAPI, LangGraph, MCP, Redis, psycopg, ClickHouse client, Azure Identity.

Assumptions used in this document

Assumption	Reason
Release title uses package version v0.1.0.	pyproject.toml declares version 0.1.0; tags are phase-oriented rather than semver release objects.
Change summaries are inferred.	GitHub tags expose tag names, dates, commit IDs and short commit messages; the detailed code diff was not available through the browser view.
Readiness scores are analytical estimates, not formal audit results.	Scores synthesize public README, tag and package metadata evidence.
Commercial-grade means document design, structure and governance transparency.	The PDF is a professional sample release note and not an official project release publication.

Source URLs reviewed

URL
https://github.com/aveeshek/bosgenesis-mop-creation-agent
https://github.com/aveeshek/bosgenesis-mop-creation-agent/tags
https://raw.githubusercontent.com/aveeshek/bosgenesis-mop-creation-agent/main/pyproject.toml
https://raw.githubusercontent.com/aveeshek/bosgenesis-mop-creation-agent/main/SPEC.md
https://raw.githubusercontent.com/aveeshek/bosgenesis-mop-creation-agent/main/TECH_STACK.md
https://raw.githubusercontent.com/aveeshek/bosgenesis-mop-creation-agent/main/PROJECT_STRUCTURE.md